

Substrate Independent Thinking Hypothesis (SITH): Life, Meaning, and the Autocatalytic Closure of Information

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Abstract

Recent work in theoretical biology and complex systems has advanced a substrate-independent conception of life, grounded not in Earth-specific chemistry but in universal principles of optimization, physical constraint, and information propagation. In particular, research on biological scaling and informational efficiency has emphasized that life is best understood as a dynamical process operating under energetic and informational limits, rather than as a specific material configuration.

This paper extends that perspective by proposing the *Substrate Independent Thinking Hypothesis* (SITH): the claim that any sufficiently closed, recursively self-maintaining organization of information constitutes a living system in a formal sense, regardless of whether it is instantiated biologically, cognitively, culturally, or technologically. Under this hypothesis, collections of documents, symbolic procedures, institutional practices, and built environments may satisfy the same viability conditions as biological organisms, provided they preserve identity, regulate transformation, suppress entropy, and propagate across time.

The paper situates SITH relative to existing frameworks including autopoietic theory, autocatalytic sets, and recent scale-based approaches to life. It then formalizes the hypothesis using category theory, information geometry, and a field-theoretic model (RSVP) that decomposes semantic dynamics into coherence potential, propagation flow, and entropy density. A sequence of worked examples bridges abstract theory and material instantiation, culminating in an analysis of bounded environments—such as bureaucratic rooms and architectural spaces—as operators that stabilize collective agency.

The result is a unified framework for analyzing life, meaning, and power as manifestations of informational closure under constraint, with implications for cultural evolution, institutional design, and the empirical study of semantic systems.

1 From Universal Life to Universal Thinking

The problem of defining life in universal terms has long been constrained by its historical association with terrestrial biology. Classical definitions emphasize properties such as metabolism, cellular organization, or biochemical self-replication, thereby embedding contingent features of Earth-based life into what is meant to be a general concept. While such definitions are operationally useful within biology, they become inadequate when extended to artificial systems, cultural evolution, or the search for life beyond Earth.

Recent developments in theoretical biology and complex systems research have begun to loosen this constraint. By reframing life as a dynamical process rather than a material category, these approaches emphasize optimization, information processing, and physical constraint as primary explanatory factors. On this view, life is understood as a learning dynamic operating across scales, subject to energetic, thermodynamic, and informational limits. Specific substrates—whether DNA, neural tissue, or silicon—are treated as historically contingent realizations rather than defining features.

This shift is consequential. Once life is defined in terms of organization and dynamics rather than chemistry, it becomes possible to ask whether similar principles govern non-biological systems that nonetheless exhibit persistence, adaptation, and constraint-driven evolution. However, most existing substrate-independent accounts stop short of fully generalizing thinking itself. They successfully abstract life, but continue to treat cognition, meaning, and symbolic organization as derivative or secondary phenomena.

The central claim of this paper is that the same substrate-independent logic that applies to life applies equally to thinking. Thinking, in this sense, is not a privileged activity of brains, nor does it require consciousness or subjective experience. It is instead a class of informational dynamics characterized by recursive structure, regulated transformation, and resistance to entropic dissolution. Wherever such dynamics occur, a form of life is present in a precise organizational sense.

This motivates the Substrate Independent Thinking Hypothesis (SITH), which can be stated informally as follows: thinking systems are living systems whose primary mode of persistence is semantic rather than metabolic. Under this hypothesis, collections of documents, legal codes, rituals, diagrams, algorithms, and architectural plans may qualify as living entities insofar as they satisfy the same viability conditions that govern biological organisms. The claim is not that such systems are conscious, but that they instantiate life at the level of informational organization.

The remainder of this paper develops this claim in stages. The next section situates SITH explicitly relative to existing theoretical frameworks, clarifying both its lineage and its points of departure. Subsequent sections generalize autocatalytic closure beyond chemistry, analyze documents as semantic organisms, formalize semantic error thresholds, and develop mathematical models using category theory, information geometry, and field dynamics. Only after this formal groundwork is established does the paper turn to material instantiations, including furniture, rooms, and built environments, which are treated as operators that externalize and stabilize collective agency.

1.1 Relation to Existing Frameworks

The Substrate Independent Thinking Hypothesis does not arise in isolation. It builds directly on several established frameworks in theoretical biology and cognitive science, while extending them into domains they were not originally designed to address. Clarifying these relationships is essential both for situating the present work and for distinguishing its claims from neighboring theories.

Autopoietic theory, as developed by Maturana and Varela, characterizes living systems as networks of processes that produce and maintain the components that constitute the network itself. This emphasis on organizational closure rather than material composition aligns closely with SITH. However, classical autopoiesis remains tightly coupled to biochemical realization, and its criteria are not straight-

forwardly applicable to symbolic or institutional systems. SITH generalizes the notion of closure by abstracting away from material production and focusing instead on the recursive maintenance of informational identity.

Stuart Kauffman’s work on autocatalytic sets similarly provides a crucial precursor. Autocatalysis demonstrates how self-sustaining organization can emerge without centralized control, given sufficient combinatorial richness and appropriate constraints. SITH adopts this insight but extends it from chemical reaction networks to semantic transformation networks. Where Kauffman’s sets catalyze molecules, SITH’s systems catalyze interpretations, procedures, and inscriptions.

Recent work emphasizing scale, information, and thermodynamic efficiency in living systems further motivates the present approach. By treating life as an optimization process constrained by physics, these accounts open the door to substrate-independent formulations. SITH can be understood as an extension of this program into the domain of meaning and symbolic structure, where the relevant constraints are informational rather than metabolic.

The primary point of divergence lies in scope. Whereas existing frameworks aim to explain life as traditionally understood, SITH proposes that the same formal conditions govern a broader class of systems. This proposal is speculative, but it is not unconstrained. The subsequent sections of the paper develop explicit mathematical criteria for semantic viability and show how these criteria can, in principle, be tested.

2 Autocatalytic Closure Beyond Chemistry

The concept of autocatalytic closure occupies a central position in contemporary origin-of-life research. In its classical formulation, an autocatalytic system consists of a set of reactions in which the products of some reactions catalyze others, forming a closed network capable of sustaining itself over time. No individual reaction is sufficient to constitute life; rather, life emerges only when the network as a whole achieves closure under its own dynamics. The critical insight is that life is an organizational phenomenon, not a property of specific substances.

The Substrate Independent Thinking Hypothesis adopts this insight while extending it beyond its chemical origins. Under SITH, autocatalysis is understood not as a property of molecular reactions but as a property of transformation systems more generally. What matters is not that molecules catalyze molecules, but that elements within a system participate in transformations that collectively regenerate the system’s own structure. Once this abstraction is made, it becomes possible to speak meaningfully of autocatalytic closure in domains that involve symbols, procedures, or representations rather than chemistry.

In semantic systems, the role played by chemical reactions is taken up by meaning-preserving transformations. These transformations include acts such as interpretation, translation, citation, amendment, compilation, or procedural execution. When a collection of symbolic elements is organized such that these transformations tend to reproduce and stabilize the collection itself, an autocatalytic semantic system is formed. Closure, in this context, refers to the fact that the system’s own rules of transformation are sufficient to maintain its identity over time.

This generalization does not eliminate the need for substrates. Symbols must be written, spoken, stored, or executed somewhere. However, the substrate functions as a carrier rather than as a defining feature. The same semantic system may migrate across substrates while preserving its organizational identity, just as a biological lineage may persist despite changes in individual organisms. Substrate independence applies to the level of organization, not to the level of implementation.

A crucial consequence of this perspective is that failure modes can be analyzed independently of material realization. Just as chemical autocatalytic systems can fail when reaction rates fall below critical thresholds, semantic autocatalytic systems can fail when transformation pathways cease to regenerate the system's identity. Collapse, in both cases, is a loss of closure rather than a loss of substance.

This section established the conceptual basis for treating semantic systems as candidates for life under SITH. The next section develops this claim concretely by examining documents and document collections as paradigmatic examples of autocatalytic semantic organization.

3 Documents as Semantic Organisms

Documents are typically treated as passive containers of meaning, serving as vehicles through which information is transmitted from authors to readers. This view, while intuitive, obscures the dynamics by which documents persist, transform, and exert causal influence over time. Under SITH, documents are not merely representations; when organized into self-maintaining systems, they function as components of living semantic organisms.

A single document, considered in isolation, does not generally qualify as a living system. Like an isolated molecule, it lacks the capacity to regenerate itself under transformation. However, when documents are embedded within a structured ecology of interpretation, citation, revision, and enforcement, they participate in a network that can maintain identity across generations of carriers. Constitutions, scientific theories, technical standards, and religious canons exemplify such systems. Their persistence does not depend on the continued existence of any particular individual, but on the stability of the transformation pathways that reproduce them.

These systems exhibit behaviors closely analogous to those of biological organisms. They replicate through copying, dissemination, and institutional adoption. They undergo variation through revision, reinterpretation, and contextual application. They compete for attention, legitimacy, and resources within shared environments. They maintain boundaries through mechanisms such as canonicity, version control, accreditation, and orthodoxy. None of these processes requires intentional design at the level of the system as a whole; rather, they emerge from the interaction of local rules and constraints.

What distinguishes a living semantic system from a mere aggregation of texts is the presence of regulated transformation. In a living system, not all changes are permitted. Some transformations are recognized as preserving identity, while others are treated as corruptions or failures. This distinction may be enforced explicitly, as in legal or technical systems, or implicitly, as in linguistic or cultural practice. In either case, the regulation of admissible transformation is the mechanism by which

semantic identity is maintained.

This perspective clarifies why document systems can exert causal power. Laws constrain behavior not because individuals believe in them, but because the institutional machinery that interprets and enforces them continues to operate. Scientific paradigms guide research not because they are universally accepted, but because they structure training, publication, and evaluation. The agency of such systems resides not in minds, but in the stability of their organizational closure.

Treating documents as semantic organisms also reframes questions of preservation and decay. Archival practices, editorial standards, and institutional memory function as error-correction mechanisms, reducing the effective mutation rate of the system. Conversely, environments that privilege novelty, speed, or engagement over coherence increase mutation rates and risk crossing semantic error thresholds. These dynamics are structural rather than moral, and they can be analyzed quantitatively once the appropriate formal tools are in place.

The next section introduces one such tool by generalizing the concept of the biological error threshold to semantic systems. This generalization provides a precise criterion for the collapse of meaning and prepares the ground for the mathematical formalizations that follow.

4 The Semantic Error Threshold

A theory that treats semantic systems as living entities must account not only for their persistence but also for their failure. In biological contexts, this role is played by the concept of the error threshold, originally developed in the study of molecular evolution. The error threshold captures a fundamental limitation: information-bearing systems can sustain only a finite amount of complexity given a nonzero rate of error in replication. Beyond a critical mutation rate, selection can no longer preserve the informational signal, and the system collapses into noise.

The Substrate Independent Thinking Hypothesis generalizes this principle to semantic systems. Documents, languages, institutions, and cultural practices all propagate information through processes that introduce variation. Copying, interpretation, translation, and contextual application are never perfectly faithful. Each transformation introduces the possibility of error, ambiguity, or reinterpretation. When such variation is sufficiently constrained and corrected, identity is preserved. When it is not, semantic coherence degrades.

The semantic error threshold can therefore be understood as a boundary in transformation space. Below the threshold, the system's internal corrective mechanisms are sufficient to stabilize meaning. Above it, uncontrolled variation accumulates faster than repair can compensate, and the system loses its capacity to reproduce itself as the same system. Importantly, this collapse does not require malicious intent or conscious neglect. It is a structural consequence of the balance between informational load and transformation fidelity.

This perspective reframes a wide range of phenomena. The fragmentation of languages under conditions of rapid mixing, the breakdown of legal systems subjected to excessive amendment or inconsistent enforcement, and the collapse of scientific fields overwhelmed by unintegrated publication all exemplify semantic error threshold crossings. In each case, the failure is not primarily ethical

or psychological, but dynamical. The system has exceeded its capacity to maintain identity under transformation.

The semantic error threshold also clarifies the role of conservatism in semantic systems. Mechanisms that appear to resist change—such as precedent, ritual repetition, or strict version control—function as error-correction strategies. They reduce effective mutation rates at the cost of slowing adaptation. Whether such mechanisms are beneficial or harmful depends on environmental conditions, but their functional role is not mysterious. They exist because they enable survival under noise.

By treating collapse as a quantitative phenomenon rather than a moral one, SITH provides a framework for analyzing semantic stability without appeal to normative judgment. The question is not whether a system ought to change, but whether it can change without losing itself. This distinction becomes crucial when evaluating modern informational environments, which often maximize variation while minimizing correction.

The next subsection addresses a common objection to this line of reasoning: that semantic systems are fundamentally dependent on minds, and therefore cannot be considered substrate-independent in the same sense as biological life.

4.1 Objection and Response: Are Semantic Systems Mind-Dependent?

A natural objection to the Substrate Independent Thinking Hypothesis is that semantic systems appear to require minds as their substrates. Words must be understood, documents must be interpreted, and symbols must be recognized as meaningful. From this perspective, semantic systems would seem to be parasitic on cognitive agents rather than independent entities in their own right. If this objection holds, the claim of substrate independence would be undermined.

The force of this objection rests on a conflation of implementation with organization. While it is true that semantic systems require physical instantiation and interaction with cognitive agents, the same is true of biological systems with respect to chemistry and physics. Cells require molecules, but life is not reducible to any particular molecule. The relevant question is not whether a system requires a substrate, but whether its defining properties depend on the specific nature of that substrate.

Under SITH, substrate independence applies at the level of organizational pattern rather than material realization. A semantic system is defined by the structure of its transformation network and the constraints that regulate those transformations. Minds function as carriers and actuators within this network, but they do not define its identity. A legal code persists despite turnover among judges and citizens; a programming language persists despite changes in hardware and user base; a religious canon persists despite variation in belief and interpretation. In each case, the system's identity is maintained by institutional and procedural constraints rather than by any particular mental state.

This distinction parallels familiar cases in biology. A virus requires host cells to replicate, yet it is not identical to those cells. A symbiotic organism may depend on another species for survival, yet retains its own evolutionary trajectory. Dependence does not imply lack of identity. What matters is whether the system exhibits closure under its own transformation rules. Semantic systems do so by constraining interpretation, enforcing admissible change, and suppressing drift.

Moreover, many semantic processes now operate with minimal or no direct human involvement.

Automated document processing, algorithmic enforcement of protocols, and machine-mediated communication increasingly execute semantic transformations without conscious interpretation. These developments do not create semantic life *ex nihilo*, but they make explicit what was already implicit: that the dynamics of meaning are not exhausted by subjective experience.

The objection that semantic systems require minds therefore does not invalidate SITH. It instead highlights the role of minds as one class of substrates among others. Just as life migrated from purely chemical processes to multicellular organization, and later to nervous systems, thinking migrates across substrates as constraints permit. Substrate independence is a claim about the portability of organization, not the absence of physical realization.

Having addressed this objection, the paper now turns to the minimal conditions under which semantic closure can arise. The next section examines the lower bound of viable organization, expressed culturally as the requirement that “two or three” carriers be jointly constrained by a shared symbolic attractor.

5 Minimal Closure and Triangulation

The preceding sections have treated semantic systems at a level of abstraction that presupposes the existence of closure, correction, and persistence. It is therefore necessary to ask what minimal conditions are required for such closure to arise in the first place. In biological contexts, the emergence of life requires a sufficient density of interacting reactions to sustain autocatalysis. In semantic contexts, an analogous requirement exists: identity cannot be maintained by a single carrier operating in isolation.

A solitary inscription, however carefully preserved, remains vulnerable to uncorrected drift. Without redundancy or comparison, error cannot be detected, and variation cannot be distinguished from corruption. When two carriers interact under a shared constraint, feedback becomes possible, but stability remains limited. Divergence between interpretations cannot be adjudicated without recourse to an additional reference. The introduction of a third constrained carrier allows triangulation. Inconsistencies can be resolved, drift can be identified, and identity can be stabilized relative to a shared standard.

This transition marks the emergence of a self-correcting semantic system. What appears culturally as a gathering around a shared name or principle can be understood formally as the creation of a minimal error-correcting code. The shared symbolic attractor defines the identity basin, while the interaction among multiple carriers suppresses unbounded variation. From this point onward, the system can persist independently of any single participant.

The importance of this minimal closure condition extends beyond small-scale interactions. Institutions, disciplines, and traditions all rely on mechanisms of triangulation to stabilize meaning. Peer review, appellate courts, and cross-checking protocols perform the same function at larger scales. In each case, semantic viability depends not on unanimity, but on the presence of sufficient structured redundancy to detect and correct error.

This analysis provides a natural bridge between abstract theory and empirical practice. It explains

why isolated genius rarely suffices to create enduring systems, and why durable meaning requires collective constraint. It also anticipates the role of architectural and institutional structures, which embed triangulation into material form. Before turning to those applications, however, it is necessary to examine how semantic systems migrate across substrates in response to constraint.

6 Ontogenic Hacking as Substrate Migration

The concept of ontogenic hacking describes the capacity of living systems to circumvent the limitations imposed by their current substrates by inventing new architectures for information storage and transmission. In biological evolution, genetic inheritance constrains both the rate and fidelity of information propagation. Nervous systems emerged as a means of storing and transmitting information within a lifetime, thereby reducing dependence on genetic change. Language further externalized information, enabling transmission between individuals without reproduction. Documents extended this process across generations, allowing information to persist independently of any particular body.

Under SITH, ontogenic hacking is not merely an acceleration of evolution, but a migration of organizational closure. Each new substrate alters the balance between complexity and fidelity by changing the effective mutation rate and the available mechanisms of error correction. The semantic error threshold is not eliminated by such migrations; it is relocated. What changes is the range of viable complexity that can be sustained.

This perspective clarifies why cultural and technological systems can evolve orders of magnitude faster than biological ones while remaining stable. The stability does not arise from intrinsic robustness, but from the availability of externalized correction mechanisms. Editorial standards, institutional enforcement, and automated validation all function as semantic immune systems, suppressing variation that would otherwise destroy identity. The cost of this stability is increased dependence on infrastructure and coordination.

Ontogenic hacking also introduces new vulnerabilities. As semantic systems migrate into faster and more interconnected substrates, the potential for rapid entropy injection increases. Unregulated propagation can overwhelm correction mechanisms, leading to collapse even in systems that were previously stable. The same technologies that enable unprecedented preservation of meaning also enable unprecedented rates of distortion.

The remainder of this paper is concerned with formalizing these dynamics and examining their concrete instantiations. The next section introduces a category-theoretic framework that specifies the structural conditions under which semantic closure can be maintained. Subsequent sections translate these conditions into quantitative and field-theoretic terms, before returning to material environments as operators that embed semantic viability into physical space.

7 Category-Theoretic Formalization of SITH

The preceding sections motivate the Substrate Independent Thinking Hypothesis conceptually. To render the hypothesis precise and defensible, it is necessary to specify the structural conditions under

which a semantic system qualifies as living. Category theory provides a natural formal language for this purpose, as it abstracts away from internal constitution and focuses instead on relations, transformations, and compositional closure.

The core idea is that a semantic system can be characterized by the transformations that preserve its identity. These transformations form a structured collection whose closure properties determine whether the system persists under perturbation. Category theory allows this structure to be expressed without privileging any particular substrate or implementation.

Consider a category, denoted $\mathcal{S}$, whose objects are semantic states. A semantic state is understood as an inscription or configuration of inscriptions together with a specification of the invariants that determine identity for that system. These invariants may include logical consistency, legal validity, syntactic well-formedness, or procedural compliance, depending on the domain. Morphisms in $\mathcal{S}$ are meaning-preserving transformations: translations, revisions, interpretations, executions, or applications that respect the relevant invariants.

Composition of morphisms corresponds to sequential application of transformations. Identity morphisms correspond to transformations that leave the semantic state unchanged. The category thus encodes the space of admissible change for semantic systems.

Within $\mathcal{S}$, a candidate living semantic system is represented by a subcategory whose objects and morphisms are closed under composition and identity. Closure here is essential. It ensures that applying admissible transformations to elements of the system yields results that remain within the system. This property captures the notion of organizational closure that underlies both biological autopoiesis and autocatalytic sets.

Replication is formalized as an endofunctor $R: \mathcal{S} \rightarrow \mathcal{S}$. This functor maps each semantic object to a new instantiation of that object, such as a copy, edition, or re-enactment, and maps each morphism to a corresponding transformation between instantiations. Importantly, R need not preserve objects exactly; it may introduce controlled variation. What matters is that the image of R remains within $\mathcal{S}$ up to semantic equivalence.

Environmental perturbations are represented by a functor that may map objects outside of $\mathcal{S}$. This functor captures contextual effects such as reinterpretation by new audiences, technological change, or political pressure. A semantic system is viable if the combined action of perturbation and replication restores it to up to equivalence. Formally, this requires that the composite admits an invariant subset of objects within $\mathcal{S}$.

This condition can be interpreted as a fixed-point requirement. The system persists if there exists a class of semantic states that remain stable under repeated cycles of perturbation and repair. The existence of such a class is the categorical analogue of biological viability.

To make this abstraction concrete, consider the case of a constitutional document. The constitution itself, along with its recognized amendments and authoritative interpretations, forms an object in $\mathcal{S}$. Judicial rulings, legislative clarifications, and formal amendments constitute morphisms that preserve constitutional identity. Radical reinterpretations or extralegal actions fall outside the admissible morphism set and correspond to semantic death or replacement.

The subcategory consists of all semantic states recognized as valid instantiations of the constitution.

Replication occurs when the constitution is cited, taught, or applied in new contexts. Environmental perturbations arise from social change, political conflict, or technological development. The system remains alive so long as its internal mechanisms—courts, precedents, amendment procedures—are sufficient to absorb perturbations and restore identity. When these mechanisms fail, the constitutional system collapses or is replaced.

This example illustrates how SITH captures the life of semantic systems without invoking metaphor. The constitution lives insofar as it maintains organizational closure under constrained transformation. Category theory provides a precise language for expressing this closure, but it does not yet quantify its stability. For that, a metric notion of distance and noise is required. The next section introduces information geometry as a means of quantifying semantic identity, drift, and collapse.

8 Information Geometry of Semantic Identity

Category-theoretic structure specifies which transformations are admissible within a semantic system, but it does not quantify how close a transformed state remains to the system’s identity. To address questions of stability, drift, and collapse, a metric notion of semantic distance is required. Information geometry provides such a notion by modeling semantic states as points on a statistical manifold equipped with a natural Riemannian metric.

A semantic system can be characterized operationally by the distributions of observable outcomes it induces. These outcomes may include linguistic tokens, legal decisions, citation patterns, execution traces, or behavioral responses, depending on the domain. For a given semantic state, these observables define a probability distribution over a measurable space. Variation in meaning corresponds to variation in the parameters of this distribution.

Let Θ denote the space of parameters indexing a family of probability distributions $\mathcal{P}$. The Fisher information metric on Θ is defined by

$$g_{ij}(\theta) = \mathbb{E}_{x \sim p(\cdot|\theta)} \left[\frac{\partial \log p(x|\theta)}{\partial \theta_i} \frac{\partial \log p(x|\theta)}{\partial \theta_j} \right].$$

Semantic identity is represented not by a single point in Θ , but by a compact region $\mathcal{I}$, referred to as the identity basin. Variations within this basin are tolerated by the system’s invariants, while variations outside correspond to semantic failure. Meaning-preserving transformations correspond to trajectories that remain within $\mathcal{I}$.

To illustrate this framework concretely, consider a simple two-parameter semantic system characterized by parameters θ , where θ_1 encodes lexical choice frequency and θ_2 encodes syntactic structure frequency in a corpus of texts governed by a shared standard. Suppose the observable distribution over token sequences factorizes approximately as

$$\begin{aligned} p(x|\theta) &= p_1(x|\theta_1) p_2(x|\theta_2). \\ g_{11}(\theta) &= \mathbb{E} \left[\left(\frac{\partial \log p_1}{\partial \theta_1} \right)^2 \right], \quad g_{22}(\theta) = \mathbb{E} \left[\left(\frac{\partial \log p_2}{\partial \theta_2} \right)^2 \right]. \\ ds^2 &= g_{11}(\theta) d\theta_1^2 + g_{22}(\theta) d\theta_2^2. \end{aligned}$$

Replication introduces stochastic perturbations in parameter space. Each act of copying, interpretation, or reimplementaion induces a random displacement whose covariance encodes the effective

mutation rate of the system. Repair mechanisms, such as editorial standards or institutional enforcement, induce deterministic flows that contract distances toward the center of the identity basin. The expected evolution of semantic distance can therefore be modeled as a balance between diffusion and contraction on the manifold.

The semantic error threshold arises when diffusion dominates contraction. Let $\bar{\kappa}$ denote the average contraction factor induced by repair mechanisms, and let σ^2 denote the effective variance of the noise introduced per transformation. Viability requires that the expected Fisher distance after a transformation-and-repair cycle decrease on average. When $\bar{\kappa} < \sigma^2$, this condition fails, and trajectories exit the basin with high probability. Identity is lost.

This formulation connects semantic stability to measurable quantities. Lexical variation rates, citation divergence, amendment frequency, or protocol incompatibility can all serve as empirical proxies for parameters in $\bar{\kappa}$ and σ^2 . The Fisher metric provides a principled way to aggregate such measurements into a single notion of semantic distance. While precise estimation may be challenging in practice, the framework specifies what would need to be measured for SITH to be empirically evaluated.

The information-geometric perspective also clarifies the role of redundancy and triangulation. Multiple independent carriers reduce effective noise by averaging perturbations, thereby shrinking σ^2 . This effect increases the size of the viable identity basin and lowers the risk of collapse. The same mechanism underlies error correction in communication systems and robustness in biological populations.

Having quantified semantic identity and drift, the paper now turns to a dynamical description of how semantic systems evolve over space and time. The next section introduces a field-theoretic formulation that integrates coherence, propagation, and entropy into a unified model.

9 RSVP Field Theory of Semantic Dynamics

The categorical and information-geometric formalisms specify the structural and metric conditions for semantic viability, but they remain largely atemporal. To analyze how semantic systems evolve in space and time, it is useful to introduce a field-theoretic description that treats coherence, propagation, and entropy as coupled dynamical quantities. The Relativistic Scalar–Vector–Entropy Plenum (RSVP) provides such a description by decomposing semantic dynamics into three interacting fields defined over a semantic manifold.

Let $\mathcal{M}$ denote a smooth manifold whose points correspond to local semantic states. The topology of $\mathcal{M}$ depends on the application. In linguistic systems, $\mathcal{M}$ may be approximated as a high-dimensional embedding space; in legal or institutional systems, it may be a graph of admissible states endowed with a continuous relaxation. The precise topology is not fixed a priori, but the assumption of local smoothness is sufficient for the present analysis.

On $\mathcal{M}$, define a scalar field ϕ , interpreted as semantic coherence potential. Regions of high ϕ correspond to states that strongly attract interpretations and transformations back toward identity. Define a vector field $\mathbf{v}$, representing propagation flow, which encodes how semantic content moves through carriers,

channels, or institutions. Finally, define a scalar entropy density field S , representing local semantic disorder, ambiguity, or uncontrolled variation.

The coupled dynamics of these fields are governed by a system of partial differential equations that balance diffusion, advection, production, and dissipation. A minimal form of the RSVP equations is given by

$$\begin{aligned}\frac{\partial \Phi}{\partial t} &= D_\Phi \nabla^2 \Phi - \alpha S + \beta |\vec{v}|^2 - \gamma \Phi, \\ \frac{\partial \vec{v}}{\partial t} &= -\nabla \Phi - \delta \vec{v} + \kappa \nabla^2 \vec{v}, \\ \frac{\partial S}{\partial t} &= D_S \nabla^2 S + \mu |\nabla \vec{v}|^2 - \nu \Phi.\end{aligned}$$

These equations admit a clear interpretation. Semantic coherence diffuses locally, is degraded by entropy, is reinforced by organized propagation, and decays in the absence of maintenance. Propagation flow is driven down gradients of coherence potential, damped by frictional effects, and smoothed by diffusion. Entropy diffuses, is generated by turbulent or uncontrolled propagation, and is suppressed by coherent structure. Together, these dynamics encode the competition between stabilization and drift that characterizes semantic life.

Boundary conditions play a critical role in determining system behavior. For a closed semantic system, such as a tightly regulated institution, one may impose no-flux boundary conditions on Φ and S , ensuring that propagation and entropy do not leak across system boundaries. For open systems, such as public discourse, boundary conditions may permit inflow of entropy or outflow of propagation, modeling exposure to external perturbation. The choice of boundary conditions corresponds to architectural and institutional constraints in the material world.

Viability corresponds to the existence of attractor regions in field space where Φ remains positive and bounded, remains structured rather than turbulent, and remains below a critical threshold. These attractors correspond to the identity basins described earlier in information-geometric terms. The RSVP equations thus provide a dynamical realization of the semantic error threshold: when entropy production overwhelms coherence generation, attractors disappear and semantic identity collapses.

The field-theoretic formulation also makes explicit the role of scale. As the size of the system increases, diffusion terms become more significant, and the maintenance of coherence requires stronger coupling or more restrictive boundaries. This scaling behavior mirrors biological constraints on organism size and organizational complexity. It also anticipates the importance of architectural and infrastructural interventions, which modify boundary conditions and coupling strengths rather than internal content.

Before turning to those material instantiations, it is useful to examine an intermediate-scale example that bridges abstract field dynamics and concrete practice. The next subsection analyzes a version-controlled software repository as a semantic system governed by the same principles.

9.1 An Intermediate-Scale Worked Example: Version-Controlled Repositories

Consider a collaborative software repository managed under a distributed version-control system. The semantic content of the repository consists of source code files together with their dependency structure and execution semantics. Commits correspond to transformations of semantic state, while

pull requests and code reviews function as admissibility checks that determine whether transformations preserve identity.

In this setting, semantic coherence potential can be operationalized as test coverage, build success, or adherence to interface contracts. Propagation flow corresponds to the rate and pattern of code changes across modules and contributors. Entropy density reflects unresolved conflicts, inconsistent conventions, or failing tests. Merge policies, continuous integration pipelines, and style guides act as boundary conditions and dissipation mechanisms that suppress entropy and reinforce coherence.

Empirically, repositories exhibit error thresholds. When commit velocity exceeds the capacity of review and testing infrastructure, coherence degrades rapidly. Bugs proliferate, interfaces destabilize, and contributors lose confidence in the system. Conversely, overly restrictive policies suppress propagation and stall adaptation. Viable repositories operate within a bounded regime where propagation is energetic but constrained, mirroring the balance required for semantic life more generally.

This example demonstrates that the RSVP framework is not limited to metaphorical applications. It provides a concrete language for analyzing real systems whose behavior can be measured, simulated, and optimized. With this bridge in place, the paper now turns to material operators that enforce similar constraints through physical form.

10 Furniture as Semantic Operators

Having established a formal account of semantic viability in categorical, geometric, and field-theoretic terms, it is now possible to examine how these abstract dynamics are instantiated in material form. Furniture provides a particularly clear case. Far from being inert or merely utilitarian, furniture functions as an operator that constrains semantic transformation by shaping access, sequence, and admissibility. Under the Substrate Independent Thinking Hypothesis, furniture is best understood not as a symbol but as a mechanism for enforcing organizational closure.

Consider a filing cabinet. Its semantic function is not exhausted by its capacity to store documents. By dividing contents into drawers, folders, and labeled compartments, it induces a topology on semantic space. Documents placed within the same drawer are rendered mutually proximate, while documents in different drawers are separated by frictional barriers. Access is discretized, order is stabilized, and accidental transformation is suppressed. The cabinet thus increases semantic coherence potential locally by reducing entropy and regulating propagation flow.

This function is independent of the specific documents involved. The same cabinet can stabilize legal records, financial accounts, or scientific data. What persists is not the content but the operator that constrains transformation. In categorical terms, furniture defines admissible morphisms by restricting which transformations can be composed without explicit effort. In RSVP terms, it imposes boundary conditions that suppress uncontrolled diffusion and turbulence.

The junk drawer presents an instructive contrast. It is often described as disordered, yet it exhibits a distinct form of semantic life. Objects placed within a junk drawer are not sorted by formal category but by anticipated reuse. Their proximity is determined by shared potential future function rather than present classification. The drawer accumulates heterogeneous items whose semantic coherence

is emergent and contextual. Entropy is higher than in a filing cabinet, but it is not unbounded. The drawer remains viable so long as users can reliably retrieve objects when needed.

This distinction illustrates a form of semantic speciation. The filing cabinet corresponds to a high-coherence, low-entropy regime suited to long-term institutional memory. The junk drawer corresponds to a flexible, exploratory regime with higher entropy but greater adaptability. Both can be alive under SITH, provided they maintain identity under use. Collapse occurs only when retrieval fails systematically, signaling a crossing of the semantic error threshold.

The power of furniture becomes evident when scaled. A bureau, archive, or records office aggregates many such operators into a hierarchical structure. Authority accrues not from symbolic legitimacy alone but from control over admissible transformation pathways. To govern the drawers is to govern the future, because it determines which semantic states can be reached with minimal cost. This observation demystifies bureaucratic power by reducing it to operator control rather than intentional dominance.

The relevance of this analysis extends beyond furniture to the architectural spaces that compose it. The next section examines rooms as higher-order compositions of semantic operators.

11 Rooms as Compositions of Semantic Operators

A room is not merely a container for furniture. It is a compositional structure that integrates multiple operators into a coherent regime of constraint. Walls, doors, windows, and furnishings together define a bounded region of semantic space with regulated ingress, egress, and internal flow. In categorical terms, a room can be modeled as a product of operator functors whose combined action constrains admissible transformations more strongly than any individual component.

Rooms enforce triangulation at scale. By co-locating specific operators and excluding others, they stabilize particular forms of interaction. A courtroom, for example, composes benches, lecterns, witness stands, and barriers into a system that enforces procedural order. The arrangement constrains who may speak, when, and with what authority. Semantic coherence is not achieved through persuasion alone, but through spatially embedded rules that regulate propagation and suppress noise.

The cultural significance of “the room where it happened” reflects this structural reality. Decisions acquire durability not merely because of who participates, but because the room itself enforces closure. Entry is restricted, documentation is controlled, and outcomes are recorded and archived. The room functions as a semantic reactor in which high-impact transformations occur under tightly constrained conditions. Its power derives from its ability to concentrate coherence and limit entropy.

This logic extends naturally to the biblical injunction to divide the ark into rooms and provide it with a window. Compartmentalization enforces separation of semantic and functional domains, preventing uncontrolled interaction. Hierarchy organizes access and flow. The window provides a controlled coupling to the environment, allowing information and resources to enter without collapsing internal structure. These features are not symbolic embellishments; they are necessary conditions for viability under perturbation.

Rooms thus serve as the minimal architectural unit at which semantic life becomes materially

stabilized. They translate abstract invariants into physical constraints, making organizational closure robust to turnover among individual agents. From rooms, it is a short step to buildings, campuses, and cities. The next section examines how such built environments function as extensions of collective agency under SITH.

12 Built Environments as Agency-Extending Operators

Furniture and rooms instantiate semantic operators at small scales, but the same principles govern larger built environments. Homes, cities, and empires are not merely settings in which human activity occurs; they are active participants in semantic dynamics. They externalize intention, constrain transformation, and stabilize collective action over time. Under the Substrate Independent Thinking Hypothesis, built environments function as large-scale operators that extend agency beyond individual bodies and lifespans.

A home integrates multiple rooms into a coordinated system that structures daily behavior. Kitchens, bedrooms, storage spaces, and circulation paths encode expectations about activity, sequence, and priority. The refrigerator externalizes memory about food availability; the stove encodes affordances for transformation; storage spaces regulate access to tools and materials. These structures bias semantic state transitions by making some actions cheap and others costly. Agency emerges not solely from decision-making, but from the alignment of intention with environmental constraint.

At the scale of cities, these effects are amplified. Roads, zoning laws, utilities, and public institutions regulate propagation flows across populations. Hospitals concentrate semantic and material resources for health-related transformations. Schools encode curricula that reproduce knowledge across generations. Administrative buildings enforce documentation, record-keeping, and procedural continuity. Together, these elements form a distributed semantic organism whose behavior cannot be reduced to the intentions of any individual participant.

Empires extend this logic further by standardizing operators across vast territories. Uniform measurement systems, legal codes, architectural styles, and administrative procedures reduce entropy by constraining variation. Power, in this context, is the ability to impose operator consistency at scale. Resistance often takes the form of entropy injection: informal practices, alternative infrastructures, or symbolic subversion that disrupts coherence potential.

This perspective reframes historical and political analysis. Revolutions are not merely ideological shifts but operator-level reconfigurations. When existing environments can no longer suppress semantic entropy or facilitate viable propagation, collapse becomes inevitable. Conversely, durable institutions persist not because of ideological purity but because their environments continue to enforce closure.

The RSVP framework makes these claims precise. Built environments increase semantic coherence potential by embedding constraints into physical form. They regulate propagation flow by channeling movement and communication. They suppress entropy by stabilizing access, enforcing procedure, and reducing uncontrolled interaction. Viability depends on maintaining these fields within bounded regimes. When environmental constraints degrade, semantic life falters.

13 Falsifiability and Empirical Engagement

A theory that claims broad applicability must specify conditions under which it could be falsified. The Substrate Independent Thinking Hypothesis does not predict that all symbolic systems are alive, nor that all built environments succeed in stabilizing agency. Instead, it predicts specific relationships between constraint, variation, and persistence that can be empirically evaluated.

One class of predictions concerns error thresholds. SITH predicts that semantic systems will exhibit sharp transitions between stability and collapse as effective mutation rates exceed corrective capacity. These transitions should be observable as sudden losses of coherence in document systems, institutions, or collaborative projects when rates of change outpace validation mechanisms. Empirical studies of software repositories, legal systems, or online communities can test for such thresholds by correlating rates of modification with measures of functional integrity.

A second class of predictions concerns operator effects. SITH predicts that introducing or removing material constraints will measurably alter semantic dynamics. For example, reorganizing archival storage should change retrieval success rates and error propagation. Modifying architectural layouts should affect decision latency, conflict frequency, or procedural compliance. Such effects can be studied experimentally or observationally using controlled interventions and longitudinal analysis.

Quantifying RSVP fields remains challenging but not in principle impossible. Semantic coherence potential can be approximated using measures of agreement, reproducibility, or predictive success. Propagation flow can be measured through communication patterns, document circulation, or movement data. Entropy density can be estimated from variance, inconsistency, or unresolved conflict. While these proxies are imperfect, they provide a starting point for empirical engagement.

Evidence against SITH would take the form of systems that maintain identity and agency without closure, constraint, or error correction, or systems that collapse despite satisfying the formal viability conditions outlined here. To date, no such counterexamples are known, but the framework invites systematic testing rather than rhetorical persuasion.

The final section synthesizes the argument and reflects on its implications for the study of life, meaning, and power.

14 Conclusion: Life as Closure, Thinking as Survival

This paper has argued that the criteria used to define life in biological systems extend naturally, and rigorously, to semantic systems once life is understood as organizational closure under constraint rather than as a specific material realization. Building on recent substrate-independent approaches to life, the Substrate Independent Thinking Hypothesis proposes that thinking itself is a form of life, instantiated wherever information maintains identity through regulated transformation in the presence of noise.

The argument has proceeded in stages. First, it situated SITH relative to autopoietic theory, autocatalytic sets, and scale-based approaches to biological life, clarifying that the hypothesis extends these frameworks rather than rejecting them. Second, it generalized autocatalytic closure from

chemistry to semantics, showing how documents and institutions can function as living systems when they regulate admissible transformation. Third, it introduced the semantic error threshold as a universal limit on information preservation, applicable to biological, cultural, and technological domains alike.

To render these claims precise, the paper developed three complementary formalizations. Category theory was used to define semantic life in terms of closure under admissible morphisms. Information geometry provided a metric structure that quantifies semantic identity, drift, and collapse. The RSVP field theory translated these abstractions into a dynamical model of coherence, propagation, and entropy evolving over semantic manifolds. Together, these formalisms establish that SITH is not merely metaphorical, but structurally grounded.

Only after this foundation was laid did the paper turn to material instantiations. Furniture, rooms, buildings, cities, and empires were analyzed as operators that embed semantic constraints into physical space. These environments extend agency by biasing state transitions, suppressing entropy, and stabilizing collective intention across time. In this light, power becomes intelligible as control over semantic operators rather than as an intrinsic property of individuals or symbols.

The resulting framework reframes familiar phenomena. Archives become organs of memory. Bureaucracies become metabolic systems. Rooms become semantic reactors. Revolutions become phase transitions driven by the breakdown of operator viability. None of these claims requires anthropomorphism or mysticism; they follow from the same principles that govern life at every scale.

The Substrate Independent Thinking Hypothesis is necessarily ambitious, and many of its implications remain to be explored. However, its central claim is modest in a precise sense: wherever information closes upon itself under constraint and resists entropy through regulated transformation, life is present in organizational form. Thinking survives by becoming architecture.

A Mathematical Appendix

The appendix provides formal justification for several claims made in the main text, focusing on existence, stability, and scaling properties of semantic systems.

Let $\mathcal{S}$ be a semantic subcategory embedded in $\mathcal{C}$, equipped with a replication endofunctor R and an environmental perturbation functor P . Define the composite evolution operator $E = RP$. A semantic system is viable if there exists a nonempty set of objects $\{S_i\}$ such that for every S_i , there exists S_j with $S_i \approx S_j$, where $\approx$ denotes semantic equivalence.

Assume that the space of semantic states admits a metric induced by a Fisher information metric. Further assume that perturbations introduce noise with bounded variance σ_c^2 , and that internal correction induces contraction with factor λ . Then, by standard results in stochastic contraction mappings, there exists a unique invariant distribution supported on a compact subset of semantic space provided that

$$\lambda^2 < \sigma_c^2 = \frac{(1-\lambda)^2}{K},$$

Stability follows from the same contraction condition. Perturbations decay exponentially in expectation so long as the contraction dominates diffusion. Scaling analysis shows that as system size increases, effective diffusion terms scale superlinearly unless counteracted by hierarchical constraint.

This result explains why large semantic systems require architectural and institutional reinforcement to remain viable.

The free energy functional introduced in the RSVP section can be derived by analogy with variational principles in nonequilibrium thermodynamics. The chosen form penalizes entropy, rewards coherence, and enforces boundary sharpness. Under the RSVP dynamics, this functional decreases monotonically in expectation within viable regimes, providing a Lyapunov function for semantic life.

Finally, measurability follows from the operational definitions provided in the main text. While precise measurement remains an open problem, the framework specifies which quantities must be estimated and how they relate. This renders SITH falsifiable in principle, even where practical challenges remain.

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